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COMPREHENSIVE REVIEW

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Risk factors affecting the food safety risk in food business operations for risk-based inspection: A systematic review

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Abstract

Foodborne illnesses result in a high disease burden worldwide, making food safety control of food business operations (FBOs) an urgent issue. With public agencies and FBOs facing challenges in monitoring the complex food supply chain with limited resources, scientific and objective insights into those factors that are related to food safety at FBOs are needed. These factors can be used as input for risk-based inspection. We conducted a systematic review to identify and analyze risk factors affecting the FBOs' food safety risk. We used a set of predefined search strings in Scopus and Web of Science to search for scientific manuscripts published in the English language between January 1 2003 and February 1 2023. The review identified 53 relevant studies and 43 risk factors. The presence of certified personnel turned out to be the most cited factor. Nearly half of the extracted factors had only been investigated in one study. Additional challenges were identified for developing a universal ready-to-use list of factors for the building of a risk-based inspection method, such as the limitation in the applicability of identified factors in different types of FBOs, and the variability in conclusions between publications for certain factors (e.g., FBO location and inspection history), stressing the need for additional research. Future studies should also prioritize standardizing definitions and measurements, particularly regarding compliance factors. In general, the current list of factors brought forward in our review lays the groundwork for building a transparent, objective, and risk-based method for food safety inspections of FBOs.

KEYWORDS
food safety, inspections, risk factors, risk-based methods

1 | INTRODUCTION

Foodborne illnesses result in a global public health burden. In 2015, the World Health Organization reported their estimations of the global burden of foodborne illnesses, predicting that more than 600 million cases of foodborne illnesses may occur in a year, causing 420,000 deaths (WHO, 2015). Increased pressure has been placed on both the private (food businesses) and public sectors (organizations owned and operated by the government) to ensure food safety and public health. Evidence suggests that most foodborne illnesses stem from factors that are mainly

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linked to mishandling by food handlers, such as inadequate cooking, cross-contamination, and poor hygiene (Fenton et al., 2006; Hardstaff et al., 2018; McIntyre et al., 2013). To date, food business operations (FBOs) all over the world are considered responsible for ensuring the safety of the food they produce and market (WTO, 2016). In many countries, public organizations have a role in the supervision of the FBOs managing food safety, where at the same time, FBOs also supervise or check each other (Havinga, 2015). However, due to the globalization of the food supply chain, and limited resources, the capability of public and private agencies to conduct food safety control over FBOs is increasingly under pressure (Havinga, 2015). In this context, a more cost-effective strategy for food safety control is required, and there has been a global trend toward risk-based controls (RBC) (van Asselt et al., 2012).

Nowadays, considerable progress has been made in developing risk assessment frameworks and methods for RBC (van Asselt et al., 2012). According to van Asselt et al. (2012), RBC can be seen as consisting of two main steps: risk classification and risk-based surveillance. Studies and reviews have focused on the first step, risk classification, concerned with identifying and ranking hazard-food combinations, for which various methods and risk assessment models have been developed, including qualitative, quantitative, and semi-quantitative methods (Focker et al., 2023; Ross & Sumner, 2002; Van der Fels-Klerx et al., 2018). However, the risk classification alone does not directly lead to a fully optimal food safety control scheme. To further optimize food safety control, a risk-based surveillance (Step 2) system is needed to help food safety authorities to prioritize FBOs according to their food safety risk profiles, hereby targeting the FBOs with the highest risk and determining the frequency of inspections (Rothstein et al., 2006; van Asselt et al., 2012).

Food safety authorities from different countries apply their own risk-based inspection methods where stark differences exist. When observing the risk-scoring methods in four European countries (the United Kingdom, Germany, France, and the Netherlands), such differences are mainly reflected in the included factors, algorithms and—in the end—the types of FBOs considered high risk (Borraz et al., 2022). Furthermore, the development processes and algorithms for these methods are rarely transparent making it hard to evaluate them. An exception is the establishment-based risk assessment (ERA) model developed by the Canadian Food Inspection Agency (CFIA) to determine the frequency of inspections of FBOs whose development processes were well-documented and published in four articles (Racicot et al., 2018, 2019; Zanabria et al., 2018, 2019). However, a comprehensive review of the factors that contribute to food safety risk of FBOs, and thus a list of those factors that can be used in the setup of a risk-based inspection method, is not available. Hence, this study aimed to obtain insights into the factors that determine the food safety risks of FBOs, as well as any knowledge gaps through a systematic review. Our goal for this review is to identify factors related to the food safety risk of FBOs and to provide a foundation for future development of a risk-based method for food safety inspections. By synthesizing the existing literature, this review seeks to provide insight into the factors that may contribute to or mitigate food safety risk at FBOs.

2 | METHODOLOGY

A systematic review was performed to synthesize and analyze articles investigating potential factors and indicators that can impact or reflect food safety risks in FBOs. This study followed the EFSA Guidance for the application of systematic review methodology (The European Food Safety Authority [EFSA], 2010). Due to the limited information on confidence levels or uncertainty measures, as well as the variability in outcome measures and examined factors of the included articles, we only evaluated the methodology quality of included studies from the study design.

2.1 | Search strategy

2.1.1 | Data sources

In February 2023, two bibliographic databases, Scopus and Web of Science, were systematically searched with a predefined search strategy to retrieve relevant scientific articles.

2.1.2 | Search strings

The search strings used in databases were developed as four groups of key terms and combined with Boolean operators "AND" and "OR" (Table 1).

The first group defined the scope of targeted results as in the food and feed safety field (i.e., "feed safety" OR "food safety"). The other three groups were developed based on three main concepts from the research question, namely, food business operator, factor, and food safety risk.

Based on Article 3(2) of Regulation No, 178/2002, we defined an FBO as "a food company, premise or any undertaking, that carries out any of the activities related to any stage of production, processing and distribution of food and is subject to official food safety inspections" (EC). Here, an FBO refers to a wide range of entities involved in food such as restaurants, food trucks, catering ser-

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TABLE 1 Search strings.

Topic	Search string
Field	"feed safety" OR "food safety"
Food business operators	"food business operator*" OR "food business*" OR "food company*" OR "food establishment*" OR "food industry" OR "food factor*" OR "food producer*" OR "food processor*" OR "food premise*" OR "food retail*" OR "food service*" OR restaurant* OR "food manufacture*" OR "primary produce*" OR "food distribution*"
Factors	"variable*" OR "factor*" OR "characteristic*" OR "practice*" OR "food safety culture" OR "food safety environment" OR "food safety knowledge" OR "risk perception*" OR "historical data" OR "compliance history" OR "compliance record*" OR "hazard-food combination*" OR "producer type" OR "type of operation" OR "processing step*" OR "type of commodity" OR "consumer exposure" OR "vulnerable population" OR "company size" OR "food safety management system*" OR "third-party audit" OR "third-party inspect*" OR "certification*" OR "certified manager*" OR "food safety personnel*" OR "preventive control*" OR "cleaning system*" OR "tracking system*" OR "train*" OR "hygienic design*" OR "risk management" OR "pest control" OR "waste management"
Food safety risk	"food safety risk*" OR "risk rank*" OR "risk score*" OR "violation*" OR "inspection*" OR "level of compliance" OR "compliance behavior?" OR "noncompliance*" OR "food safety breach*" OR "food safety performance" OR "foodborne illness outbreak*" OR "foodborne illness risk*" OR "foodborne illness incident*" OR "foodborne disease outbreak*" OR "foodborne disease risk*" OR "foodborne disease incident*" OR "contamination incident*" OR "contamination risk*"

vices, food manufacturers, processors, and distributors. Factors refer to any conditions or characteristics related to or taken by FBOs that can impact food safety risks (e.g., "compliance record*," "company size," and "food safety management system [FSMS]"). Terms reflecting FBOs' food safety risk were based on three main perspectives, namely, food safety compliance (e.g., "inspection*" and "violation*"), microbiological, and chemical hazards (e.g., "foodborne illness outbreak*," "foodborne disease [FBD] outbreak*," and "contamination incident*"), and food safety performance.

The search focused on scientific articles published in the English language, which were the main languages used by our research team. We included peer-reviewed articles that were published within the last 20 years (January 1 2003–February 1 2023), considering that the start of the period risk-based regulation was introduced and became more prevalent.

2.2 | Study selection

All initially retrieved results were stored in an Endnote database and duplicates were removed. The remaining results were then evaluated and classified through three tiers.

Tier 1: The titles, keywords, and abstracts of the results were reviewed to determine the type of article and to classify the articles into three main categories, relevant, possibly relevant, and not relevant, based on the predefined eligibility criteria.

Tier 2: The results classified as relevant were read in full. Possibly relevant references were further filtered by reading the full text and those that met the eligibility criteria were classified as relevant, whereas the rest were grouped into not relevant.

Tier 3: References of relevant articles were snowballed to obtain possibly relevant results that were missed and then assessed again into Tiers 1 and 2.

Based on the EFSA guideline, we adopted a sequential review method for the data collection, where one reviewer collected and grouped the data, and another reviewer verified the results through an inter-rater check on a randomly selected subset (5%) of all results grouped after going through Tiers 2 and 3. Any disagreements were resolved through discussion. The final consistency rates were recorded.

2.3 | Eligibility criteria

We targeted studies that explored the correlation between risk factors and food safety risk of FBOs and saved these studies as *A1 studies*. In addition to A1 studies, other articles presented information about factors adopted by official food safety authorities to decide their inspection frequencies or predict food safety compliance of FBOs (such as a review article discussing risk factors adopted by countries) were also included as *A2 articles*. With such an approach, we aimed to ensure a comprehensive coverage of factors that are included in scientific articles but not implemented in practice (A1) and vice versa factors implemented in risk-based inspection but not necessarily

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included in scientific articles (A2). Eligibility criteria were established for these two types of articles.

For A1 articles, studies that examined factors that are directly applicable or applicable after some modifications in a risk-based inspection method were included. We excluded studies that examined factors that require on-site inspection of handling practices as these are not known upfront and can thus not be used as input to determine inspection frequency. Furthermore, the results of such on-site inspections are automatically included when discussing historical inspection reports.

In addition, only studies examining the correlation between factors and food safety risk were included. Studies examining the correlation between a factor and an indirect indicator for food safety risk, such as food safety motivation of food handlers, or food safety practices were excluded.

For A2 articles, no other limitations on study designs were set; any peer-reviewed articles containing information about factors used for deciding inspection frequency or predicting FBOs' food safety compliance were included.

2.4 | Data processing and analysis

Retrieved relevant A1 and A2 articles were reviewed, and the following information from the presented study was extracted from the articles into Excel forms:

For A1 articles: Author (year); main purpose; country/region; study design; type of FBO; participants; sampling method; sample size; factors; indicator of factor/intervention; outcome of interest; outcome measure tool; outcome indicator; result.

For A2 articles: Author(year); title; country/region; type of food business; factors; indicators; weights of factors; frequency of inspection.

Factors were saved into different groups during the extraction of data. Based on CFIA's categorization of food safety risk factors, we classified the studied factors into three main categories (Racicot et al., 2019). The first group is "Inherent factors," including factors related to the characteristics of the FBOs, such as size or location. Factors referring to measures or certifications applied by the FBOs that aim at decreasing food safety risks were grouped as "Mitigating factors." For the last group, factors that can be used to predict or affect FBOs' compliance behavior were summarized as "Compliance factors." Each category is further divided, with inherent factors detailing FBO type, size, location, and facility and structure, mitigating factors including training and certification, food safety management and certification, and composite index of food safety practices, and compliance factors mainly covering historical data and food handlers' compliance behavior drivers.

The retrieved results were summarized and analyzed in a qualitative manner, where the study design (e.g., observational cohort study, observational cross-sectional study), significance of results (whether a significant difference was found), number of evidence on each factor, and the consistency of conclusions were discussed for a better idea of the relevant factors and the applicability of factors in assessing food safety risks.

3 | RESULTS

3.1 | Study selection

The outcomes of the systematic review are summarized in Figure 1. Initially, 564 records were retrieved from Scopus and 357 from Web of Science. After removing duplicates, 610 records were left to enter the 3-tier screening process. From Tier 3, five potentially relevant records were identified through snowballing references from the post-Tier 2 relevant results. These were also included in Tier 1 for screening, resulting in 615 records. Of these, 428 articles were considered not relevant in Tier 1 (screening on title, keywords, and abstract). Full-text reading resulted in another 132 nonrelevant articles ultimately leading to 560 nonrelevant articles. In total, 53 relevant articles were found: 44 for A1, 13 for A2, including 4 articles that did not only meet the requirements for A1 articles but also contained information required for A2 articles, which were thus credited to both. The consistency rate of the inter-rater check was 77.4%, and disagreements were solved through discussion.

3.2 | Study characteristics

3.2.1 | A1 articles

A total of 44 eligible A1 articles were included, published between 2004 and 2023, from research groups in the United States ($n = 28$), Brazil ($n = 4$), China ($n = 3$), Canada ($n = 2$), Saudi Arabia ($n = 1$), the United Kingdom ($n = 1$), Finland ($n = 1$), Lebanon ($n = 1$), Belgium ($n = 1$), Dubai ($n = 1$), and Germany ($n = 1$). An interesting trend observed was that only one relevant article per year was published prior to 2011, whereas most relevant studies were published in the last decade, with a maximum of seven relevant articles published in 2020.

Despite our aim to include all FBO types when designing the search protocol, only three of the included articles indi-

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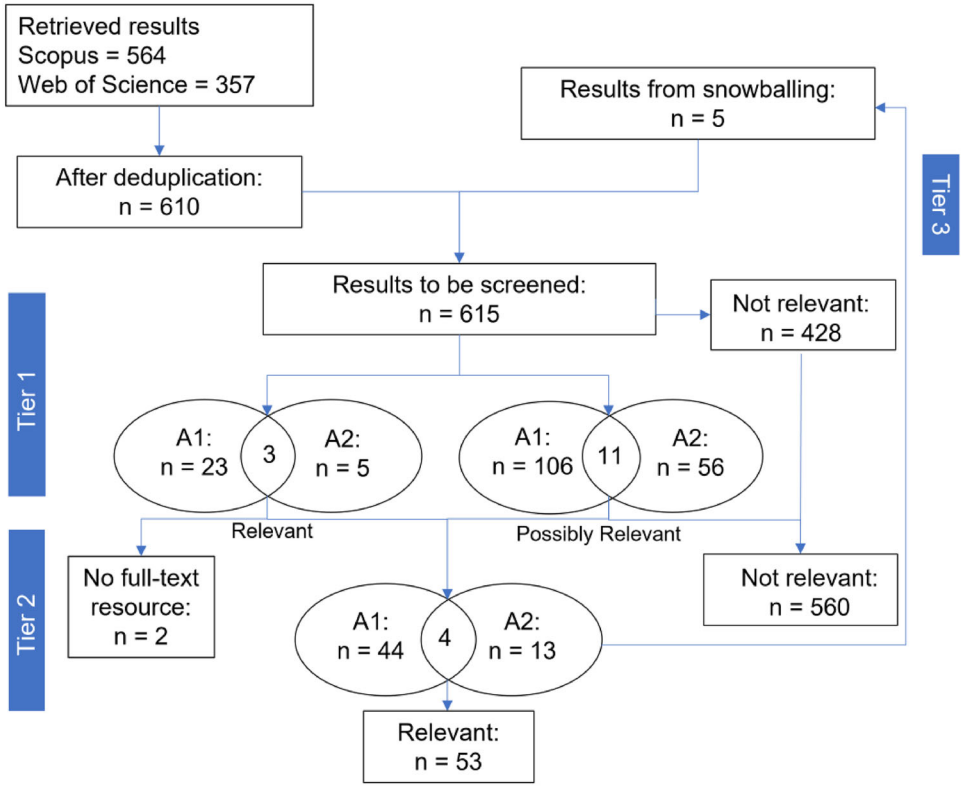


FIGURE 1 Results of study selection process.

cated that their study population included food premises or food companies, whereas the majority of A1 studies ($n = 37$) focused exclusively on food service establishments where the food is distributed or served to consumers (e.g., restaurants and institutional catering), with a large proportion of which focused solely on restaurants ($n = 23$).

The studies retrieved measured the FBOs' food safety risk level based on three main aspects, namely, noncompliance risk ($n = 29$), food safety performance ($n = 9$), and foodborne outbreak risk ($n = 7$) (one measured both food safety performance and noncompliance risk). Most studies focusing on noncompliance risk used inspection results ($n = 25$). In this case, the number of critical violations was widely used, which is defined as violations about time and temperature, hygiene practices by food handlers, cross-contamination, and food from unapproved sources (Kassa et al., 2010). The other four studies that focused on noncompliance risks adopted surveys and checklists, together with third-party audits and internal audit results, which were also used by all studies assessing food safety performance. Foodborne outbreak risk was estimated using foodborne outbreak reports ($n = 6$) or using laboratory analysis results ($n = 1$).

As for the food safety factors examined, 25 articles studied inherent factors, 16 articles mitigating factors, and 13 articles examined compliance factors. The description of the factors, the grouping, and the types of FBOs studied for these factors are presented in Table 2. More detailed data are provided in Table S1.

3.2.2 | A2 articles

Among the 13 included A2 articles, 11 of them were published between 2018 and 2022, one in 2003, and one in 2010. Four articles were also included in A1, as these provided information about risk factors used in the United States and Canada in the introductions of the studies (Appling et al., 2018; Barros et al., 2020; Jemaneh et al., 2018; Kassa et al., 2010). As shown in Table 2, unlike the A1 studies, almost half of the A2 studies targeted general FBOs ($n = 6$), and five of them focused on food service establishments, leaving two articles that provided no information on the type of FBOs examined. In addition, four review articles provided more detailed and comprehensive factors used and gave a better explanation of the risk-based methods used in several countries (Borraz et al., 2022; Martin et al., 2003; van Asselt et al., 2021; Wang, Li et al., 2022). The remaining five studies explored new risk-based ranking methods. Three articles were written within the same project of building a risk assessment model for food establishments in Canada (Racicot et al., 2018, 2019; Zanabria et al., 2018). Due to the overlap of these three articles, they were

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TABLE 2 Factors, description of the factors, the grouping, and the types of food business operations (FBOs) studied for the factors.

Factor	Subgroup		A1[a] FBO[b]	A2[c] FBO
Inherent factor	FBO type	General		
		Establishment/service type	FSE[d]	FSE
		Business operation type	FSE	n/a[e]
		Commodity/food/cuisine type	FSE	All FBOs
		Activity of the business	n/a	All FBOs
		Integration level	FP[f]	n/a
		Price	FSE	n/a
		Processing/preparation/handling complexity type	FSE	FBOs
	FBO size	Company size	FSE	n/a
		Number of outlets	FSE	n/a
		Production volume	n/a	NI[g]
		Revenue	FSE	n/a
		No. of workers/labor cost	FSE	n/a
		Consumer population	n/a	FBOs
	FBO location	Spatial characteristic	FSE	FSE
	FBO facility and structure	*Adequacy of structures and facilities such as walls, doors, floors, ceilings, area for hand hygiene, equipment, furniture and utensils*	FSE	FBOs
Mitigating factor	Training and certification	Training system/program	FSE	NI
		Certified food safety personnel on site	FBOs; FSE	FBOs
	Food safety management and certification	General		
		Implementation of FSMS	n/a	FBOs
		Certification status of FSMS	FSE	n/a
		Implementation of an International scheme certification	n/a	FBOs
		Use of HACCP	n/a	NI
		Certified self-checking system (SCS)	FP	n/a
		Sampling and controls	n/a	FBOs
		Third-party audits	FSE	FBOs
		Personnel management	n/a	FBOs
		Traceability/documentation of product chains	n/a	NI
		Additional processes to further reduce food safety risk	n/a	FBOs
	Composite index of food safety practices	The intensity of food safety practices	FP	n/a
		Management practices	FP	n/a
Compliance factor	Historical data	Inspection results	FSE	FBOs
		Risk category	n/a	FBOs
		Foodborne illness/outbreak history	n/a	FBOs
		History of enforcement actions	n/a	FBOs
		Number and type of recalls	n/a	FBOs
		Number of food safety complaints	FSE	FBOs
		Hygiene status	n/a	FBOs

(Continues)

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TABLE 2 (Continued)

Factor	Subgroup		A1[a] FBO[b]	A2[c] FBO
	Food handlers' compliance behavior drivers	Material and nonmaterial behavioral drivers	FBOs	n/a
		Employee outlook	FSE	n/a
		Psychological empowerment climate	FSE	n/a
		Employee commitment	FP	n/a
		Food safety knowledge	FSE	n/a
		Food safety culture/climate	FBOs; FSE	n/a
		Confidence in management/ control procedures	n/a	NI

Abbreviation: FSMS, food safety management system.
[a]Articles that examined the relationship between risk factors and food safety risk or performance of food business operators.
[b]Type of food business operator targeted in articles.
[c]Articles containing information about factors currently used in defining official inspection frequencies or predict food safety compliance of food business operators.
[d]Food service establishments.
[e]Not applicable, no relevant study.
[f]Food processors.
[g]No information.

considered simultaneously, and factors were extracted from Zanabria et al. (2018). One article proposed a new risk-scoring and categorization method for Brazil (Viera et al., 2022). Another study, Oldroyd et al. (2021), explored a machine learning approach to predict food safety compliance. To conclude, based on the 13 articles included as A2, factors used in 14 countries were extracted and grouped into a factor list together with all the extracted factors from A1 articles (Table 2) and will be discussed in the next sections.

3.3 | Inherent factors

We identified four types of food safety risk factors that are related to the inherent characteristics of FBOs and grouped into one of the following four subgroups: FBO type, FBO size, FBO location, and FBO facility and structure.

3.3.1 | FBO type

More than one third of the A1 articles explored the relationship between FBO type and FBOs' food safety risk from different perspectives ($n = 19$). Most of the studies found FBO type a significant factor related to food safety risk, making FBO type a relevant factor to be included in a risk ranking model. The factor, FBO type, was further subdivided into service type, operation type, and others.

3.3.1.1 Establishment type

Five studies examined different establishment types, which mainly include the three types of institutional service establishments (hospital cafeterias, school catering, workplace cafeterias, and similar establishments), commercial restaurants, and street food.

Three studies found that institutional service establishments have either better food safety performance or fewer recorded violations than commercial restaurants which in turn outperformed street food (da Cunha et al., 2015, 2018; Kassa et al., 2010). Such results can be attributed to the fact that institutional service establishments tend to be better equipped to address the challenges associated with the implementation of a comprehensive FSMS, such as the substantial costs and the needed technical expertise (Mensah & Julien, 2011). Many commercial restaurants and street food vendors can only afford to employ lower wage, part-time workers, often having lower or limited educational backgrounds (Cavalli & Salay, 2007). Similarly, a recent study in Saudi Arabia attributed their finding that hotels had a lower incidence of food safety risk situations than restaurants due to the better implementation of international standards in their FSMS (Almansouri et al., 2023). Noticeably, Erickson (2015) found no significant difference between the inspection scores of mobile vendors and restaurants, and some of the street vendors even had better scores than restaurants. Erickson believes that compared to restaurants, consumers' proximity to food preparation gives street vendors more incentives to comply with food

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safety regulations, and fewer items on the menus and fewer customers help mobile vendors to further reduce violations (Cates et al., 2009; Erickson, 2015; Seiver & Hatfield, 2000).

Three studies from the United States focused on the type of service in restaurants yielding varying results. Two of them compared fast food and full-service restaurants. Of these, Cates et al. found significantly more critical violations in full-service restaurants as compared to fast food services, whereas Jones et al. (2004) failed to find any significant difference. Menachemi et al. (2012) classified the service type into more groups and showed that quick service tends to have more critical violations than cafeteria or diner restaurants. The difference in their conclusions could be attributed to the different food safety regulations and policies applied as they targeted restaurants from different states in the United States.

Overall, studies on establishment type (i.e., institutional restaurant and street vendor) appear to be more consistent than service type (i.e., fast food and full-service restaurant), where institutional service establishments may need less attention from food safety inspections.

From A2 articles, we only identified one study in the United States stating that, in Ohio, the function of food establishments was taken into account when determining the inspection frequency of FBOs (Jemaneh et al., 2018).

3.3.1.2 Operation type

Seven studies explored the relationship between FBOs' operation type and food safety compliance, all focusing on restaurants. The main types of operations examined were chain/independent operations and ethnic/nonethnic operations.

As the only study that looked at both operation types, Roberts et al. (2011) found significantly more violations in independent ethnic restaurants than in independent nonethnic restaurants, chain nonethnic restaurants, and chain ethnic restaurants. In addition, two studies found that nonethnic restaurants outperformed ethnic-operated counterparts (Harris et al., 2015; Roberts et al., 2011). The reason for this discrepancy may stem from the lack of comprehension among restaurant operators regarding local regulations and health inspectors' expectations (Harris et al., 2015). Similar to Roberts et al. (2011), when comparing the total number of violations, Leinwand et al. (2017) found that chain restaurants cited fewer violations than non-chain restaurants. Four other studies also showed that chain restaurants tend to have fewer critical violations than independent ones (Harris, 2017; Harris et al., 2014; Murphy et al., 2011; Roberts et al., 2011). This is likely due to the fact that chain restaurants have more resources available (Phillips et al., 2006) to invest in training and equipment (Frash et al., 2004), and implement control programs and standards (Roberts et al., 2005). However, in one study, although significantly fewer critical violations were found in chain restaurants compared to independent restaurants, the difference became nonsignificant when it came to total violations (Murphy et al., 2011). One possible explanation for this could be that chain restaurants tend to focus on reducing critical violations rather than other violations. Interestingly, an opposite relationship was reported by Barros et al. (2020), with independent operations showing lower odds of inspection failure when compared to chain operations (Barros et al., 2020). Barros et al. (2020) attributed this contradictory finding to the inclusion of other significant predictor variables, such as cuisine type and certification, in their multivariable models, which revealed a distorting effect on the relationship between chain establishment and inspection outcomes (Barros et al., 2020).

In summary, the type of restaurant operation, whether chain or non-chain, can be used to distinguish restaurants with different noncompliance risk levels. The A2 articles did not report on this factor.

3.3.1.3 Others

We summarized five other factors related to FBO type, including food or commodity type, processing complexity type, business activities (e.g., slaughtering, processing, packaging, and warehousing), integration level, and average entrée price.

According to the A2 articles, three factors have been widely used by various countries, being food or commodity type (the United Kingdom, Germany, Canada, and Australia), processing complexity type (France, Brazil, Canada, Australia, the United States, and China), and activity of the business (France, Canada, the United States, Australia, the United Kingdom). However, unlike A2 articles, we only found three A1 studies about food or commodity type and only one article about processing complexity type (Barros et al., 2020; Gormley et al., 2012; Jones & Angulo, 2006; Rickamer Hoover et al., 2020). Although two studies reported a significant difference among different food types of restaurants, no general conclusion can be drawn since one (Gormley et al., 2012) grouped food types as cuisines from different countries, whereas the other one grouped them as cuisines from different continents (Barros et al., 2020).

Rickamer Hoover et al. (2020) examined the processing complexity type and found that foodborne outbreaks tend to last shorter for less complex restaurants. The reason behind this can be that a more complex process increases the probability of cross-contamination, and more complex food items may also be stored and served for longer duration, leading to extended exposure (Rickamer Hoover et al., 2020). More studies on the processing complexity type are needed to support this preliminary conclusion and the use

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of this factor to distinguish between high- and low-risk FBOs.

One study looked at the average restaurant entrée price and found significantly fewer violations in restaurants with an average entrée price of $20 or more (Burke et al., 2014). However, the applicability of entrée price in deciding food safety inspection frequency is questionable, as the calculation of average entrée price was not provided in the article, and the comparability of food prices between regions and countries remains unclear. For the integration level, two studies found significantly lower food safety risk in FBOs that have higher integration levels. However, they had defined the integration level differently. Wang et al. (2022) focused on tea firms in China and defined the level of backward integration as the percentage of fresh leaves harvested in their own tea plantation. On the other hand, Abebe (2020) included multiple aspects such as information technology connection and logistic integration in their assessment of the integration level of FBOs, which is not only about the scale but also other aspects. It is difficult to draw conclusions about the impact of the level of integration in the absence of a common measure.

In general, processing complexity type, business activities, integration level, and average entrée price require more evidence before being used to assess the food safety risk of FBOs.

3.3.2 | FBO size

A2 studies about Croatia, Denmark, and China referred to "company size" as one of the risk factors when designing inspection plans (van Asselt et al., 2021; Wang et al., 2022). However, the definitions or indicators of "company size" were not always provided in the respective articles. Sha et al. (2020) based their categorization of food establishment sizes on the criteria established by the Gansu Food and Drug Administration (2014), which classifies these establishments according to their physical room size in square meters. Meanwhile, according to EU recommendation 2003/361, the size of enterprises can be defined with financial indicators such as turnover (the total amount of sales you make over a set period) and balance sheet data. We, therefore, summarized factors extracted from A1 and A2 articles that reflect both financial size and physical size into one group "FBO size." Eight A1 articles and eight countries are included in this section.

Considering the FBO size from a financial perspective, we identified three relevant A1 articles, but they reported conflicting results. Mathe (2012) found that restaurants' food safety scores were higher when they had higher revenues and lower labor costs, whereas Cates et al. (2009) observed more critical violations in establishments with higher revenues (i.e., annual revenues >$500,000) compared to those that had lower revenues. Such associations can be explained in different ways, as the revenue amount can be affected by many issues. For instance, Cates et al. (2009) believed that establishments with higher revenues serve more customers and are thus more likely to have violations. However, Mathe (2012) interpreted higher revenue as gaining more resources to promote learning food safety practices. Therefore, based on the conflicting results, further research is needed to explore the reliability of using the revenue to assess the food safety risk of FBOs.

Two studies have investigated the physical size of FBOs. One study measured actual area size and discovered that larger restaurants have a higher food safety risk, whereas another study counted the number of outlets and found a smaller number of violations in restaurants with more outlets (Kassa et al., 2010; Sha et al., 2020). In general, a larger area size being related to a higher food safety risk can be explained by the fact that these restaurants tend to have more complex operations and serve more customers. On the other hand, more outlets can also make such restaurants more experienced in food safety management which assures a better food safety level. FBO size can also be characterized by the number of workers in the FBO as was examined by Rickamer Hoover et al. (2020). However, this factor was not significantly associated with outbreak size or duration.

Based on the studies classified as A1, although the FBO size is relevant to food safety risks, finding a general and effective indicator for it is difficult due to the varying ways of expressing FBO size. From A2 articles, we found two factors related to FBO size that were used by multiple studies, namely, production volume and consumer population (Appling et al., 2018; Barros et al., 2020; Borraz et al., 2022; Jemaneh et al., 2018; Kassa et al., 2010; Racicot et al., 2018, 2019; Zanabria et al., 2018). However, no A1 studies exploring these two indicators were found.

3.3.3 | FBO location

Four of the A1 articles examined the relationship between regional or spatial characteristics of the FBO location and food safety risks, focusing on different variables.

Two studies set in the United States both found that restaurants in the Miami area tend to have relatively more violations than other regions studied (Harris et al., 2014; Lee et al., 2019). However, the possible reasons behind this difference were not examined (Harris et al., 2014). Lee et al. (2019) looked into the beach and airport proximity and found significantly more food safety violations in hotel restaurants closer to the beaches, and those further away from the airports. They attributed the situation

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to the regional difference in food safety culture and administrative systems, but further research is needed to explore the possible explanation. In the United Kingdom, FBOs' neighborhood was included as a regional characteristic to predict food safety compliance (Oldroyd et al., 2021).

The other two articles turned their attention to the socioeconomic characteristics of the regions (Ng et al., 2020; Sha et al., 2020). The level of economic development was found to negatively affect the food risk score, and a correlation was observed between higher population densities and an increased number of food safety violations (Sha et al., 2020). It was explained in the study that low-income regions tend to have a low level of education, which could mean that the food handlers and operators have poor food safety risk literacy thus leading to a high food safety risk (Lam et al., 2013). In addition to economic characteristics, language is another regional characteristic that gained interest. In Canada, Ng et al. (2020) found a correlation between the number of critical food safety violations and the percentage of immigrants, as well as the percentage of non-English/French native speakers in the neighborhood of the restaurants. Regions with more immigrants are more likely to face the language barrier, which makes it hard for them to comply with the regulations.

In general, it is shown that the FBO location is related to the food safety risk level of restaurants. However, the significant findings of the included studies can hardly lead to a unified conclusion as they focused on different variables. Countries and cities may need their own studies to observe and derive a trend, as the situation can vary among different regions. Studies are also needed on the effect of socioeconomic factors of the region on the restaurants located there.

3.3.4 | FBO facility and structure

According to A2 studies, the adequacy (proper size and condition) of FBO architectural structures and facilities—such as walls, doors, floors, ceilings, as well as area for hand hygiene, equipment, furniture and utensils—has been used as a risk-scoring parameter in studies from Germany, Brazil, and China (Borraz et al., 2022; Viera et al., 2022; Wang et al., 2022). We identified one A1 study reporting that an adequate physical structure (i.e., proper size and condition of the physical structure (walls, doors, floors, ceilings)) can reduce the number of violations (de Andrade et al., 2021). Despite the limited number of relevant studies, it is important to note that the design and layout of facilities and structures have been included as an essential part of FSMSs, such as HACCP (U.S. Food and Drug Administration, 1997).

3.4 | Mitigating factor

Measures or factors that are considered to be able to reduce food safety risks in FBOs were grouped as "Mitigating factors." We identified two main types of these factors: training and certification, and food safety management and certification. In addition, an extra group was added for two studies that constructed and examined their own indexes for multiple indicators relevant to the adoption of food safety practices (composite index of food safety practices).

3.4.1 | Training and certification

For the application of training and relevant certifications, we identified 12 A1 studies, of which 10 focused on the presence of certified food safety personnel, and 3 studies examined the effect of training programs (1 overlap).

Certified food safety personnel on site

Nine of the 10 studies were performed in the United States. They referred to certified food safety personnel as Certified Food Safety Managers (CFSM) or Certified Kitchen Managers (CKF). Their conclusions varied depending on the measurements of the outcome. When looking at the number of critical violations (Cates et al., 2009; Kassa et al., 2010), observed noncompliance (Appling et al., 2018, 2019), and the outbreak risk (Hedberg et al., 2006; Rickamer Hoover et al., 2020), six studies found better food safety in restaurants with CFSM than those without CFSM on site. The improved food safety knowledge level of managers from the training for certification was considered the main reason behind the results (Binkley et al., 2008). However, when comparing the general inspection results (i.e., inspection score or the total number of violations) of restaurants with or without CFSMs, no significant differences were found (Binkley et al., 2008; Kassa et al., 2010). Harris (2017) even found that restaurants with CFSM had higher number of violations cited during inspections than those without CFSM, which is unexpected. They attributed the inconsistency to the possibly different effects of CFSMs on different food safety risk items as CFSMs may tend to put more attention on critical violations than on other violation items. Jemaneh et al. (2018) tried to find the evidence for the conjecture by examining the effect of CFSMs on six different aspects of food safety violation items, namely, food from unsafe sources, poor personal hygiene, inadequate cooking, improper holding temperature, contaminated equipment, and other violations but only found significantly lower number of other violations in restaurants with CFSM. Further research is still necessary to

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explore the reason why the presence of CFSM cannot lead to the improvement of overall inspection results while decreasing the number of critical violations.

Only study was conducted outside of the United States, namely, in Canada. In this study, premises with Certified Food Handlers were found significantly less likely to fail inspections. According to A2 articles, Canada has been the only country that took "certified food safety personnel onsite" into account when deciding inspection frequency.

Training program

The effect of training programs was examined by three A1 studies, with two of them reporting that restaurants that attended training programs have significantly less outbreak risk and violations (Averett et al., 2011; Rickamer Hoover et al., 2020). One study only observed an improvement in the inspection results of restaurants participating in the SAFE program (a food safety training program emphasizing high-risk food practices from receiving to service) but failed to find a significant difference (Ripley et al., 2021). Similar to the certified food safety personnel discussed earlier, the result can be attributed to the emphasis on high-risk food risks of the SAFE program.

To sum up, training and certification are shown to play an important role in improving the food safety level of restaurants, although they do not always show a significant effect on inspection results. Training and certification can be seen as an effective risk factor when ranking the FBOs' food safety risk (Barros et al., 2020; Borraz et al., 2022; Racicot et al., 2018, 2019; Zanabria et al., 2018).

3.4.2 | Food safety management and certification

When it comes to food safety management and relevant certifications, we identified 10 factors from 4 A1 articles and 6 A2 articles, such as the implementation of FSMS, certification status of FSMS, or third-party audits.

The effect of third-party audits on the inspection results has been examined by two studies, and significantly better results were observed for establishments that adopted third-party audits (Appling et al., 2018, 2019). Based on this study, Canada has decided to include third-party audits as one of their risk factors (Racicot et al., 2019).

As for the certification of FSMS, Piira et al. (2021) reported an inconsistent influence on compliance among different food establishment types and inspected items. Therefore, the adoption of a certified FSMS cannot support a decrease in food safety inspection. Similarly, according to Jacxsens et al. (2015), there was no significant difference in food safety output between companies that implemented a certified self-checking system (SCS) and those that did not. However, from the cluster analysis, these authors concluded that SCS-certified companies have a more robust food safety output, making the presence of SCS a potential checking point for risk ranking.

It is worth noting that—as shown in the factor list (Table 2)—we identified a few more items used by countries in their risk-scoring methods of FBOs which were not examined by any of the A1 studies, pointing the way to future research. For example, aside from the third-party audits, Canada also took the adoption of FSMS, sampling control plans, and some food processing methods (e.g., steam pasteurization and modified atmosphere packaging) into consideration for their risk-scoring methods aiming at deciding the food safety inspection frequency of FBOs (Barros et al., 2020; Racicot et al., 2018, 2019; Zanabria et al., 2018). FBOs adopting the documentation of product chains, HACCP, or their own sampling control plans can also gain a decrease in their inspection frequency in Germany, although as a matter of fact, the HACCP system has been a legal requirement in the EU since 2004, making it a redundant factor for FBOs in the EU (Borraz et al., 2022; EU Regulation (EC) No 852/2004). These factors can potentially be included for assessing the food safety risk of FBOs.

3.4.3 | Composite index of food safety practices

Two A1 studies established composite indexes of food safety practices. Both studies found a positive correlation between their composite index and FBO food safety performance (Abebe, 2020; Taha et al., 2020). The detailed items included in their indexes are provided in Table S2. Although we cannot draw conclusions due to the different items included in the indexes, the idea of creating a composite index to include the relevant food safety management items previously discussed could be an effective solution when building a risk-ranking model in the future.

3.5 | Compliance factor

In this section, we discuss factors related to a food handlers' compliance behavior, for which we identified factors related to historical data and the drivers of compliance.

3.5.1 | Historical data

Only five A1 studies examined the relationship between historical data and food safety risk or performance of FBOs. Four out of these five examined the relationship

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between historical inspection results and foodborne outbreak risk, and the remaining one examined historical complaint reports and the number of violations.

When examining the historical complaints report, Jemaneh et al. (2018) used the number of priority violations as the outcome of interest. They adopted the definition of priority violations provided by The District of Columbia Food Code, who identified the violation of certain provisions as priority violations, that is "provisions whose application contributes directly to the elimination, prevention, or reduction to an acceptable level of hazards associated with foodborne illness" and found a positive correlation between the number of foodborne illness complaints and two of the priority violations, being improper holding temperatures and contaminated equipment (DCDOH, 2012). Based on the finding from Jemaneh et al. (2018), the CFIA has selected the complaints record as one of their risk factors in their risk assessment tool (Racicot et al., 2018, 2019; Zanabria et al., 2018).

When focusing on historical inspection reports, in two studies, a larger number of critical violations and violations relevant to protection from contamination were found in the restaurants in which an outbreak took place than in restaurants where no outbreak took place (Firestone et al., 2020; Petran et al., 2012). However, they did not examine the significance of the difference, leaving the reliability of using historical violation number for predicting the foodborne outbreak unclear (Firestone et al., 2020; Petran et al., 2012). Rickamer Hoover et al. (2020) looked into the characteristics of restaurants experienced a norovirus outbreak and found that the presence of one or more critical violations in the most recent inspection before the outbreak significantly predicted smaller outbreaks (i.e., less suspected and confirmed primary illness cases associated with the outbreak than average). Following Lee and Hedberg (2016), such inconsistencies may stem from the different types of foodborne outbreak examined. Another possible explanation is that restaurants are required to take corrective measures after receiving citations for violations during inspections, and such prompt actions can help mitigate the potential impact of outbreaks. Meanwhile, as not all the FBOs take appropriate actions immediately after getting cited violations, the former explanation of the finding from Rickamer Hoover et al. (2020) still requires further research. Yeager et al. (2013) took the most recent two inspection results of restaurants into account and compared those who consistently scored lower in both inspections (those did not take appropriate actions after first inspection) with those that scored higher but found no difference in bacterial counts between foods from the two groups of restaurants.

In a nutshell, considering the contradicting findings about the historical inspection data, it is clear that only one or two records of historical inspection cannot be used as a reliable factor to reflect the food safety risk of FBOs, as the inspection itself works not only as an assessment of food safety risk but also as a measure to help FBOs decrease the risk by remediating their food safety practices when cited serious violations. Therefore, instead of the recent inspection results, studies should focus more on the long-term inspection data to explore the applicability of using historical inspection data for the design of risk-based inspection plans.

Based on A2 studies, at least seven countries have considered the historical inspection results in their risk-scoring methods, namely, the United Kingdom, Germany, the Netherlands, Ireland, Portugal, Malta, and Canada (Barros et al., 2020; Borraz et al., 2022; Racicot et al., 2018, 2019; van Asselt et al., 2021; Zanabria et al., 2018).

The numbers of violations and complaint reports have been adopted by many countries, as well as other types of historical data for the risk-scoring of FBOs, such as risk category (risk categorization of nonconformities for the different programs assessed, e.g., sanitation, employee hygiene, pest control, recall, equipment maintenance, water supply) (Canada), foodborne outbreak history (Canada), history of enforcement actions (Canada, China), number and type of recalls (Canada), and historical hygiene status (Germany, Brazil, China) (Borraz et al., 2022; Racicot et al., 2018, 2019; Viera et al., 2022; Wang, Wang et al., 2022; Zanabria et al., 2018). To conclude, more studies are needed to validate the reliability of using different types of historical data for the design of inspection plans.

3.5.2 | Food handlers' compliance behavior drivers

Factors that may motivate food handlers in FBOs to comply with food safety regulations and improve food safety performance are included in this section. In the eight relevant A1 studies, behavior drivers related to both workplace and employees were discussed.

Three out of the four studies that examined workplace-related factors assessed the effect of food safety culture on FBOs' food safety risks respectively from three perspectives, namely, FBD risk scores, inspection results, and food safety performance score (de Andrade et al., 2020, 2021; Jhuang & Chen, 2019). Results of each of the three studies support the fact that a better food safety culture can lead to better food safety in FBOs. However, there is a lack of harmony in the definition and measurements of food safety culture. Of the three studies, two assessed normative beliefs, work pressure, communication, self-commitment, environmental support, management, and coworker support, and leadership for food safety climate

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using a five-point Likert scale (de Andrade et al., 2020, 2021). Alternatively, Jhuang and Chen (2019) referred to a former study about food safety culture for their measurements, the Jespersen maturity model, which assesses organizational culture, food science, and social cognitive science (Jespersen et al., 2017; Jhuang & Chen, 2019). In addition to what has been reviewed, there are many other definitions of food safety culture in use, making it difficult to generalize and explore their reliability in determining inspection frequency (Jhuang & Chen, 2019). A unified definition and assessment tool is urgently needed to explore the practical use of food safety culture in a risk ranking tool.

Instead of examining the general food safety climate of FBOs, Roseman et al. (2017) focused on the psychological empowerment climate, as a variable that can be changed to improve the social problem (Roseman et al., 2017; Spreitzer, 1996). They found a positive association between psychological empowerment climate and food safety performance. The reason for the greater food safety performance can be that employees in FBOs with a high psychological empowerment climate tend to respond quickly to potential food safety problems, and their good attitudes and behaviors can motivate other employees in the FBO (Roseman et al., 2017; Wallace et al., 2011).

When it comes to behavioral drivers related to employees, Fietz et al. (2018) divided the drivers of their study into two groups: material and nonmaterial behavioral determinants. Material behavior determinants include: perceived detection risk, lack of time, costs of compliance, combination of lack of time and costs of compliance, and reputational damage. Nonmaterial behavior determinants include embarrassment, acceptance of rules, and feelings of guilt. Among all these determinants, it was found that a feeling of embarrassment, if others were to learn about noncompliance, is positively related to compliance results. This finding does not support its use as a risk indicator by official authorities, as embarrassment cannot be measured prior to inspection. However, it does suggest that transparency could be the key to more effective regulation (Fietz et al., 2018).

The indirect impact of food handlers' food safety knowledge on the food safety compliance of FBOs has been extensively examined by many studies, mainly focused on the change in food handlers' food safety practices (Banna et al., 2022; KwoI et al., 2020; Lee et al., 2017). However, when we limited the inclusion criteria of A1 studies to a direct impact on food safety, only one study explored the relationship between food safety knowledge and inspection results, failing to demonstrate a correlation (Burke et al., 2014). The result is consistent with Lee et al. (2017), who observed that although food handlers had adequate food safety knowledge, it was not translated into practice, which, in the end, could lead to no correlation between food safety knowledge and inspection results.

Interestingly, the two A1 studies about compliance behavior drivers can be seen as follow-up research, in which the effect of employee commitment and employee outlook on the transfer of knowledge gained from training to real action had been explored (Frash & Maclaurin, 2010; Taha et al., 2020). Results from Taha et al. (2020) indicated that employee commitment positively and significantly influenced FBOs' food safety performance and played a significant role in the relationship between training and food safety performance. Regarding employment outlook, that is, employee perceptions of different items that may influence food-handling behavior, Frash and MacLaurin (2010) conducted a survey that measured employee's perceptions of the company's training, self-efficacy, social responsibility, facility design, organizational culture, and health department relations and found that self-efficacy and social responsibility have positive relationships with restaurant health inspection scores. Similarly, they attributed this connection to the positive influence of employee' self-efficacy and social responsibility on the transformation of food safety training into practical food safety practices, as supported by prior studies (Clayton & Griffith, 2008; Holton et al., 2000). Future investigation on mediating factors that help translate food safety knowledge into actual improvements in food safety outcomes is merited.

It is not surprising that no country was found in A2 studies to have adopted any of the discussed factors in this section, as most of them cannot be easily measured. Only the United Kingdom was mentioned to be using a risk factor that can be grouped here: confidence in management or control procedures, though its definition and measurement method were not described (Borraz et al., 2022).

In general, despite the evidence proving their effectiveness on food safety results, compliance behavioral drivers discussed can hardly be included in risk ranking methods due to the lack of a unified definition and measurement tools. Notably, the British Retail Consortium has adopted one of the various measurement models of food safety culture built by Taylor and Rostron (2018) as an addition to their annual food safety audit (van Asselt et al., 2021). However, the evaluation is mainly based on self-assessment, and inspectors may find it too complicated to use, thus hard to be adopted by the official authorities.

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4 | DISCUSSION

This systematic review summarized and analyzed information about 43 risk factors that can impact the food safety risk of FBOs.

4.1 | Innovations

A former review provides an overview of risk-based inspection approaches in Europe, including factors that can be used to assess the risk of an FBO not complying with food safety laws or regulations (van Asselt et al., 2021). Three main elements stood out in this review, namely, company size, historical data, and compliance behavior. However, the focus on Europe and a restrictive definition of risk, defined as "risk of noncompliance," limited the comprehensive inclusion of risk factors (van Asselt et al., 2021). When it comes to the CFIA, they broadened the definition of risk into the likelihood of the occurrence of a food safety issue and retrieved a total of 155 risk factors which were later refined and simplified into 17 risk factors through expert elicitation (Racicot et al., 2019; Zanabria et al., 2018). It should be noted that the factor identification process of the ERA model was conducted in a manner where once the factor was found in one article, all subsequent articles examining the same factor were not retained. In our study, we conducted a systematic review that includes and analyzes all relevant articles to identify and evaluate risk factors. An innovative point of our study lies in the inclusion of not only studies exploring the relationship between factors and food safety risk of FBOs (A1) but also articles that provided information on factors used by countries for risk-based inspections (A2) for a more comprehensive factor list. This approach led to the identification of an additional 16 factors from A2 studies, which had not been examined in any A1 studies.

4.2 | Limitations and implications for future study

The difference between the factors (additional 16 factors) found from A2 studies relative to the A1 studies can be explained by three possible reasons. Five of the factors included in A2 studies are obligatory in some countries and are common-sense elements, such as the implementation of FSMS, the implementation of an international scheme certification, the use of HACCP, sampling and controls, and documentation of product chains. Six of the additional A2 factors lacked clear definitions and partially overlapped with A1 factors, including personnel management, additional processes to further reduce food safety risk, history of enforcement actions, hygiene status, and confidence in management and control procedures. The remaining five factors require further study to ascertain their relationships with food safety risks of FBOs (i.e., production volume; consumer population; risk category; foodborne illness and outbreak history; number and type of recalls).

Considering the reliability and applicability of examined factors for future risk-based inspection methods, it is worth discussing the study design and settings of A1 studies. All of the A1 studies are observational studies, which is mainly a choice determined by the intricacies of food safety risk, where numerous control factors make experimental research impractical. Furthermore, the vast majority of them were cross-sectional studies ($n = 39$), with only four cohort studies, leaves the causal link between most factors and food safety risk unclear. Four cohort studies were exclusively carried out on mitigating factors, more specifically, on training programs and certifications. The distinct natures of the three different groups of factors identified in this article, that is, inherent factors, mitigating factors, and compliance factors, explain the limited number of cohort studies performed: Inherent and compliance factors are closely tied to FBOs and tend to stay stable over time, making it impractical for a cohort study. In contrast, mitigating factors can be adopted at a particular time, allowing for a comparison of food safety risks before and after their implementation. Therefore, aside from training programs, more cohort studies can be done on other mitigating factors in the future.

Various countries were included in the articles studied. Notably, the majority of A1 studies were conducted in developed countries (39 out of 44), among which the United States accounts for almost three fourths ($n = 28$). The remaining five studies were carried out in developing countries, with three set in China, one in Lebanon, and one in Saudi Arabia. Considering the diverse socioeconomic conditions, policy frameworks, and cultural contexts, findings and factors derived from studies conducted in developed countries may not readily apply to the circumstances found in developing countries, and conversely. It is important to extend the scope of investigations to encompass both developed and developing nations. Future research in this field should also prioritize investigations conducted within developing countries to further explore the applicability of factors examined in developed countries, such as mitigating factors and historical food safety data which have rarely been investigated in developing countries, as well as identify their unique challenges and additional factors.

In addition to the geographical limitation, there is a notable constraint related to the business types of FBOs

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investigated. As stated earlier in Section 3.2.1, the majority of the articles included in our review focused on food service establishment (e.g., restaurants). This emphasis raises a significant concern as it may limit the broader applicability of the identified 19 factors based on restaurants only. For example, the average entrée price is a factor that can only be adopted for the risk-based inspection of restaurants. The same applies to the five factors extracted from four studies that focused on food processors. Table 2 provides information regarding which FBOs were studied for each factor. There is a necessity for follow-up research to investigate the relevance of these factors in other FBO contexts. Furthermore, there are certain factors, such as establishment type and price of a meal, that may exclusively apply to food service establishments and thus cannot be included in universal risk ranking methods directly in the future (WHO, 2024). An optional factor list for specific business types of FBOs can be a potential solution.

When it comes to the outcome measure for food safety risk, we found that the majority of studies ($n = 25$) utilized compliance results assessing food safety risks which aligns with the further goal of deciding inspection frequency. Despite the diversity in the measurements—spanning noncompliance risk, food safety performance, and foodborne outbreak risk—the consistent results across studies for most factors underscore their reliability in evaluating food safety. Nonetheless, it is important to notice that conflicting conclusions can arise from differing outcome measures. For instance, as mentioned before, discrepancies in results across studies about the correlation between historical inspection data and foodborne outbreaks were attributed to the variations in the specific outbreak types considered in the outcome measures (Lee & Hedberg, 2016). Meanwhile, although all three measures adopted—noncompliance risk, food safety performance, and foodborne outbreak risk—consider both microbiological and chemical risks, research mainly focused on mitigating microbiological risks. It is important for future studies to explore in more depth factors associated with chemical risks in FBOs.

Within the three main groups of factors, most information was found for the inherent factors, particularly in relation to FBO type. Among them, the establishment type and business operation type have been proven to significantly influence the food safety risk within food service establishments by a few studies. We also noticed that there are interactions between some of the factors, for example, the type of food and operation type (especially ethic and non-ethic) can affect each other. Future studies should look into these interactions to better understand the combined effects of these factors on food safety risks.

Based on the studies performed by the CFIA, we included factors relevant to training and certifications and food safety management as mitigating factors (Racicot et al., 2018, 2019; Zanabria et al., 2018, 2019). According to the CFIA, the impact of mitigating factors in food safety risk was highly scored by experts (Racicot et al., 2019). We found consistently significant results from 14 studies when it comes to training and certifications. However, within food safety management factors, the third-party audit can be seen as a relatively reliable risk factor, with two studies both finding positive results, whereas researchers failed to find significant results when examining the impact of the presence of SCS and the certification status of FSMS on food safety risks. Meanwhile, seven more food safety management factors extracted from A2 studies can be considered for future research and risk ranking methods.

Compliance factors included five A1 studies on historical data and eight studies on compliance behavior drivers. Among the compliance factors, an unexpected inconsistency in the results of historical inspection data suggests that, unlike long-term trends, short-term historical inspection results may lack reliability as an indicator of food safety risk. Although not examined in A1 studies, other types of historical data, such as foodborne outbreak history and the number of recalls, mentioned in A2 studies are all potential risk factors worth exploring in the future.

From the six factors related to compliance behavior drivers, our systematic review revealed that three of them—material and nonmaterial behavioral drivers, food safety knowledge, and employee commitment—require more evidence of their impact on food safety risks, whereas food safety culture is proved to be a relevant factor by three studies (Burke et al., 2014; Fietz et al., 2018; Taha et al., 2020). However, a significant limitation surfaces concerning the assessments and a variety of definitions of these factors related to compliance behavior drivers across different studies. For instance, when evaluating employee outlook, the measurement incorporated organizational culture, potentially overlapping with the concept of food safety culture (Frash & Maclaurin, 2010). In the studies about food safety culture, different measurements and definitions were adopted (de Andrade et al., 2020, 2021; Jhuang & Chen, 2019). This inconsistency extends to factors in other groups, such as FBO location, with studies employing diverse measurements for neighborhood sociodemographic data and region characteristics (Harris et al., 2014; Lee et al., 2019; Ng et al., 2020; Sha et al., 2020). To address this limitation, future research may benefit from establishing a universally agreed-upon definition and standardized measurements for these factors. Moreover, the adoption of a composite index, as demonstrated in the mitigating practices section, could offer a solution when considering behavior drivers in risk ranking, consolidating multiple indicators for a more comprehensive assessment.

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5 | CONCLUSION

Through our systematic review, we identified 53 relevant studies, from which we extracted 15 inherent factors (subdivided in FBO type, FBO size, FBO location, FBO facility, and structure), 14 mitigating factors (i.e., training and FSMSs and their certification), and 14 compliance factors (i.e., historical data and food handlers' compliance behavior drivers) that are related to the food safety risk of FBOs. The presence of certified personnel gained the most attention, with 10 studies specifically examining its impact, making it a relatively more reliable factor in comparison to others.

The relationship between most of the factors and FBOs' food safety risk requires more research, as almost half of the factors were only examined by one study, and some studies came to contradictory conclusions about factors, such as certification of FSMS, FBO location, and historical inspection results. The focus on food service establishments raises concerns for future research about the general applicability of identified factors to the other FBOs, especially inherent factors. Decision-makers should also be aware that some promising factors can only be applied to a certain type of FBO, such as restaurants, for which designing a factor list with optional factors for specific types of FBOs can be a solution.

In addition, inconsistencies in definitions and ways of measurement across studies for some factors, especially compliance factors, stress the need for standardized assessments. In a follow-up study, these could be further explored. Through identifying and analyzing a comprehensive risk factor list that may affect FBOs' food safety risks, our research can be used as the first step in building a transparent and objective risk-based inspection method to optimize resources in the future.

AUTHOR CONTRIBUTIONS

Y. Ze: Writing—original draft; funding acquisition; writing—review and editing; methodology; resources; data curation; conceptualization; investigation; software; visualization; project administration; formal analysis. **E. D. van Asselt**: Writing—review and editing; supervision; methodology; validation; data curation; conceptualization; project administration; formal analysis. **M. Focker**: Writing—review and editing; supervision; validation; conceptualization; formal analysis; project administration. **H. J. van der Fels-Klerx**: Funding acquisition; writing—review and editing; supervision; project administration; conceptualization; formal analysis.

CONFLICTS OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

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